Radiotherapy for vaginal cancer

Radiotherapy uses high-energy rays called radiation to destroy cancer cells. You may have radiotherapy as a treatment for vaginal cancer.

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What is radiotherapy?

Radiotherapy uses high-energy rays called radiation to treat cancer. It destroys cancer cells in the area where the radiotherapy is given.

When radiotherapy is used

You may have radiotherapy as a treatment to cure vaginal cancer. You may have it instead of surgery, if a general anaesthetic or surgery is not suitable for you. Radiotherapy can also be given after surgery if the cancer comes back in the same area (recurrence).

Adjuvant radiotherapy

Your cancer specialist may talk to you about having adjuvant radiotherapy after surgery for vaginal cancer. This is to reduce the risk of the cancer coming back in the area. Your specialist team will discuss the type, stage and grade of vaginal cancer to decide if adjuvant treatment might be suitable for you.

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Palliative radiotherapy

Radiotherapy can be given to help control symptoms, such as bleeding. It may also be given when the cancer has spread in the pelvic area, or to further away parts of the body such as the bones. This is called palliative radiotherapy.

How is radiotherapy given?

There are 2 ways of having radiotherapy for vaginal cancer:

- External beam radiotherapy is given from outside the body (externally) using a radiotherapy machine.
- Internal radiotherapy is when a radioactive material is placed inside the body. It is called brachytherapy.

Chemoradiation

Sometimes radiotherapy is given with chemotherapy. This is called chemoradiation. The chemotherapy makes the cancer cells more sensitive to radiotherapy. This can help radiotherapy work better.

The chemotherapy drug used is called cisplatin. You usually have it once a week during external beam radiotherapy treatment.

External beam radiotherapy for vaginal cancer

External beam radiotherapy is when a radiotherapy machine aims high-energy rays at the area of the body being treated.

This treatment is normally given as a number of short, daily treatments in a radiotherapy department. These are called treatment sessions or fractions. It usually takes about 5 to 6 weeks to have the full course of treatment. The radiotherapy machine looks like a large x-ray machine or CT scanner. There are different types of radiotherapy machine. The most commonly used machine is called a linear accelerator (LINAC).

Internal radiotherapy (brachytherapy) for vaginal cancer

Internal radiotherapy (brachytherapy) is usually used to give an extra dose of radiation to the tumour after external radiotherapy. Though some people with smaller, early-stage vaginal cancer may have brachytherapy alone.

There are 2 different ways to give vaginal brachytherapy:

Intracavity brachytherapy

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Before treatment, you will have a radiotherapy planning appointment. During this appointment you will have a vaginal examination. The radiographer will then put a plastic treatment applicator in your vagina. The end of the applicator will have lubricating gel. You might find the feeling of pressure at the top of the vagina uncomfortable, but it should not be painful. You can ask for painkillers if this would make you more comfortable. Or a gel containing local anaesthetic can be used if needed.

Once the applicator is in place, the radiographer will lay your legs flat on the bed and you will have a CT scan or MRI scan. After this, the applicator is removed and you can go home.

During treatment sessions, you will have the applicator placed in your vagina. The end that is outside your body is connected to the brachytherapy machine. The radiographer will leave the room but they can still see you and talk to you via an intercom. If the applicator is uncomfortable tell the radiographer. You can ask for painkillers if you feel that would help. Treatment takes around 15 minutes. After treatment, the applicator is removed and you can go home.

Interstitial brachytherapy

Depending on the size and location of the cancer, you may have a slightly different type of internal brachytherapy. This is called interstitial brachytherapy. You will have to stay in hospital for this type of brachytherapy.

Before treatment, you will either have a general anaesthetic or a spinal anaesthetic. A general anaesthetic is when you are asleep. A spinal anaesthetic is where you are unable to feel anything from the waist down. If you have a spinal anesthetic, you will be given sedation to make you feel relaxed and sleepy

The doctor will insert a treatment applicator into your vagina, and narrow tubes into the cancer. They will also put a small, plastic tube into your bladder (catheter) to keep your bladder empty. The applicator and catheter will stay in while you are in hospital. Once the applicators are in place, you will have an MRI scan or CT scan to check the position.

After having the applicator inserted, you will be on the ward, or in the radiotherapy department. You will stay in bed the whole time the applicator is in place. If the applicator is uncomfortable, tell the radiographer as they can give you regular painkillers to help.

During the treatment session, the brachytherapy is given through the applicator. The machine is operated by a radiographer or a medical physics technician. The machine places a radioactive capsule into the applicator. This is called the source. The machine then gives the planned dose of radiation. When the treatment is finished, the radioactive source goes back into the machine.

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Early menopause and radiotherapy

If you still have periods, radiotherapy to your pelvis can affect your ovaries and cause your periods to stop. This is called the menopause. This can cause menopausal symptoms such as hot flushes and vaginal dryness. Going through menopause also means you can no longer get pregnant.

If you have not been through the menopause and would want to have a future pregnancy, talk to your doctor. They can talk to you about having surgery to move your ovaries. The surgery would move your ovaries out of the area of radiotherapy treatment. This surgery does not always protect the ovaries. Radiotherapy to the womb area can also affect your chances of getting pregnant.

We have more information about how cancer treatment can affect fertility.

If treatment has caused early menopause and menopausal symptoms, your doctor can talk to you about the hormone replacement therapy (HRT) which may help.

Side effects of pelvic radiotherapy

Side effects do not usually happen straight away. They may develop during treatment or in the days or weeks after treatment finishes. Sometimes side effects get worse for a time during and after you have finished radiotherapy before they get better. There may also be a risk of side effects that are long-term or that only start months or years after radiotherapy.

Your team will explain what to expect. Always tell them if you have side effects during or after radiotherapy. They can give advice and support to help you cope.

When you have radiotherapy to the vagina and surrounding areas it is known as pelvic radiotherapy. We have more information about the side effects in our general information about pelvic radiotherapy.

Side effects of radiotherapy for vaginal cancer

These are some of the possible side effects of radiotherapy for vaginal cancer.

Tiredness

Radiotherapy often makes people feel tired. Tiredness (fatigue) may get worse as the treatment goes on. If you are having chemoradiation you may feel more tired.

Bladder changes

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Pelvic radiotherapy can irritate the bladder. You may feel like you need to pass urine (pee) more often or feel you cannot wait. You may also have a burning feeling when you pee. Tell your radiographer or specialist nurse if you have bladder symptoms or there is blood in your pee. They can suggest things to help.

Bowel changes

Pelvic radiotherapy can irritate your bowel. It can cause loose stools (poo) or diarrhoea, or sometimes hard stools (constipation). You may have cramping pains in your tummy or lower bowel (rectum) or pass a lot of wind. Tell your radiographer or specialist nurse if you have bowel symptoms or there is blood in your poo. They may suggest making some changes to your diet during radiotherapy. Or they may suggest treatments to help. Follow their advice carefully. Tell them if your side effects get worse. Bowel side effects usually start to improve about 2 weeks after radiotherapy finishes.

Effects on the vagina

You may have a small amount of vaginal discharge after treatment has finished. If it continues or becomes heavy, tell your cancer doctor or specialist nurse.

Pelvic radiotherapy can also cause vaginal dryness. This can continue after treatment ends. This can feel uncomfortable, particularly during sex. There are lots of products you can try such as lubricants and vaginal moisturisers. You can buy them at a pharmacy or online, or your doctor can prescribe them.

Pubic hair

When you have external beam radiotherapy, you may lose some of your pubic hair. After treatment, it usually grows back but it may be thinner than it was before.

Skin changes

External beam radiotherapy can cause the skin in the area being treated to become red or darker. It might feel sore or itchy. Tell your radiographer or specialist nurse if you have this.

Changes in your blood

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External beam radiotherapy can reduce the number of blood cells your body makes. This is a common side effect of chemotherapy, so it is more likely to happen if you are having chemoradiation.

Late effects of pelvic radiotherapy

Most people have side effects during pelvic radiotherapy. Usually these gradually improve over a few weeks or months after treatment finishes. Late effects of pelvic radiotherapy may:

- affect you during or shortly after treatment
- last longer than 3 months
- not affect you during treatment but begin months or even years later
- be a delayed response to treatment.

Common late effects include:

- changes to how the bladder or bowel work
- changes to your sex life
- tiredness.

It is always important to tell your GP or cancer doctor about any new symptoms that develop a long time after treatment. They need to be checked as they may not be caused by radiotherapy.

Late effects of radiotherapy for vaginal cancer

These are some of the other possible late effects of vaginal cancer treatment.

Bowel and bladder changes

After pelvic radiotherapy, you may develop changes to the bowel or bladder. It is common to have some mild changes, but much less common to have severe side effects that affect your quality of life. If this happens, symptoms may develop months, or sometimes years, after radiotherapy treatment.

Sex

Pelvic radiotherapy can cause changes that affect your sex life. These difficulties may be temporary and gradually improve over time. But even permanent changes that affect your sex life can often be improved. Pelvic radiotherapy may cause:

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- reduced desire for sex or a low sex drive (libido)
- changes in sexual sensation
- menopausal symptoms
- vaginal changes, such as dryness, narrowing, bleeding or infection
- changes to the anus or rectum.

We have more information about coping with changes that affect your sexual wellbeing. This includes information about managing side effects that affect the female pelvis and hormones.

Effects on the vagina

Pelvic radiotherapy can cause changes to the vagina. It can make the walls of the vagina fragile causing bleeding. If you notice any bleeding during, or at any time after treatment, tell your cancer doctor or specialist nurse. They will examine you and explain whether it is likely to be caused by the radiotherapy.

Pelvic radiotherapy can also make your vagina narrower and less stretchy. The vaginal walls may be dry and thin and can stick together. This can make vaginal sex and internal examinations uncomfortable. Your radiotherapy team may give you vaginal dilators to help stop the vagina from narrowing.

After treatment, you should be given advice on things that can help with effects on the vagina such as vaginal moisturisers and lubricants.

Lymphoedema

Pelvic radiotherapy may increase the risk of swelling in one or both legs. This is called lymphoedema. It is not common, but the risk is higher if you have surgery to remove the lymph nodes as well as radiotherapy.

Changes to the pelvic bones

Pelvic radiotherapy can cause thinning of the bones in the pelvis. This often does not have any symptoms but is seen on scans. Sometimes it may cause fractures in the pelvis called pelvic insufficiency fractures. These can cause pain in the lower back or pelvis. If this happens, your doctor can give you painkillers to help. You may also be referred to a physiotherapist.

Support from Macmillan

Macmillan is here to support you. If you would like to talk, you can:

- call the Macmillan Support Line on 0808 808 00 00.

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- chat to our specialists online.
- visit our vaginal cancer forum to talk with people who have been affected by vaginal cancer, share your experience, and ask an expert your questions.

About our information

This information has been written, revised and edited by Macmillan Cancer Support's Cancer Information Development team. It has been reviewed by expert medical and health professionals and people living with cancer.

References

Below is a sample of the sources used in our vaginal cancer information. If you would like more information about the sources we use, please contact us at informationproductionteam@macmillan.org.uk

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Our cancer information meets the PIF TICK quality mark.

This means it is easy to use, up-to-date and based on the latest evidence. Learn more about how we produce our information.

The language we use

We want everyone affected by cancer to feel our information is written for them.

We want our information to be as clear as possible. To do this, we try to:

- use plain English
- explain medical words
- use short sentences
- use illustrations to explain text
- structure the information clearly
- make sure important points are clear.

We use gender-inclusive language and talk to our readers as ‘you’ so that everyone feels included. Where clinically necessary we use the terms ‘men’ and ‘women’ or ‘male’ and ‘female’. For example, we do so when talking about parts of the body or mentioning statistics or research about who is affected.

You can read more about how we produce our information here.

How we can help

Clinical Information Nurse Specialists

Our Cancer Information Nurse Specialists are dedicated cancer nurses available to talk to on our Macmillan Cancer Support Line.

Call us free on 0808 808 0000

8am to 8pm, 7 days a week. You can talk to our cancer specialists about anything.

Email the Macmillan Support Line

Email the Macmillan Support Team if you have any questions.

Chat online

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Available 8am to 7.45pm, 7 days a week. For confidential, real-time support if you don't want to

Online Community

An anonymous network of people affected by vaginal cancer which is free to join. Share experiences, ask questions and talk to people who understand.

Find cancer support services in your area

What's going on near you? Find out about support groups, where to get information and how to get involved with Macmillan where you live.

Get help